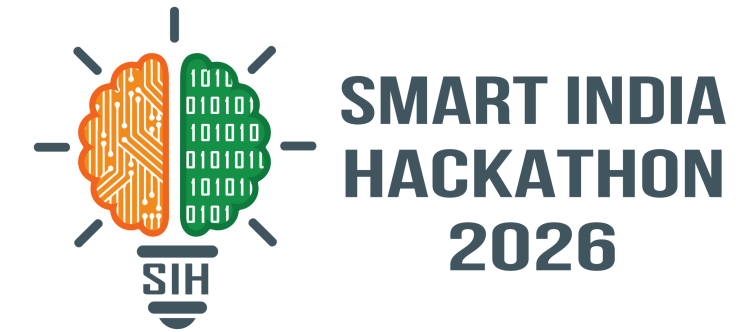


SMART INDIA HACKATHON 2026

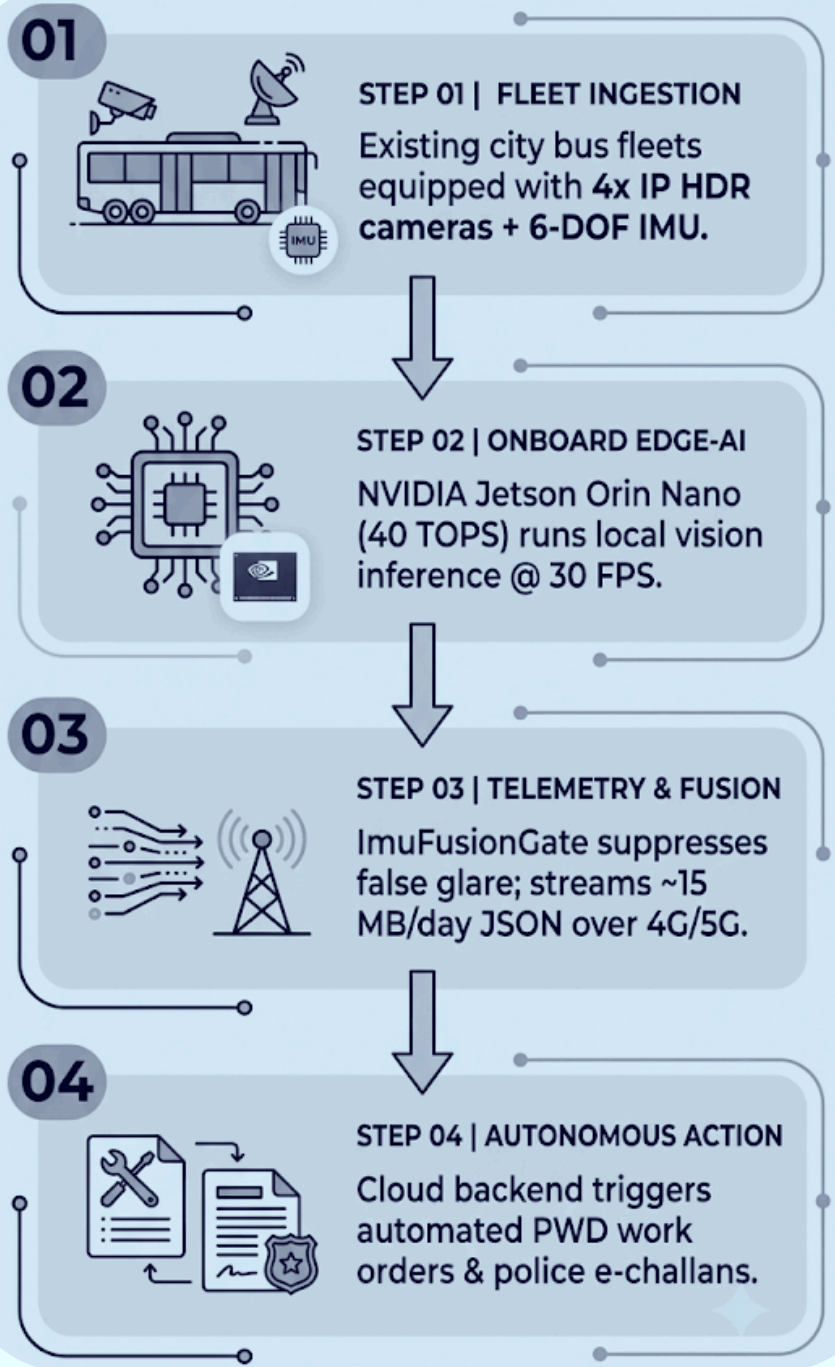
TITLE PAGE



- Problem Statement ID – SIH26124
- Problem Statement Title - AI-Powered Mobile Urban Intelligence Platform Using Public Transport Fleet
- Theme - Smart Automation
- PS Category - Software
- Team ID -
- Team Name - Arclane

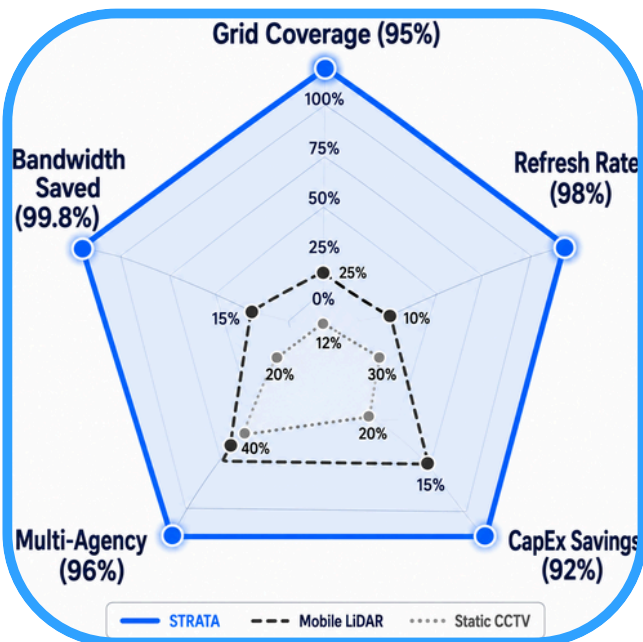


Proposed Solution



1. Detailed explanation of the proposed solution

- **Mobile Edge-AI Grid:** Turns standard city buses into autonomous inspection nodes covering 85%–95% of municipal roads every 15–30 minutes.
- **On-Bus Processing:** 40 TOPS NVIDIA Jetson edge nodes process 4-camera IP streams locally to detect potholes, waterlogging, missing medians/dividers, faded zebra crossings, and damaged traffic signboards, streaming only ~15 MB/day of encrypted JSON telemetry instead of raw video.
- **Dual-Tier Mesh:** Onboard *visual + IMU vibration* data syncs to the cloud backend for spatial clustering and *automated road ticket generation*.
- **Traffic Analytics:** *Real-time vehicle detection, classification, and density counting*.



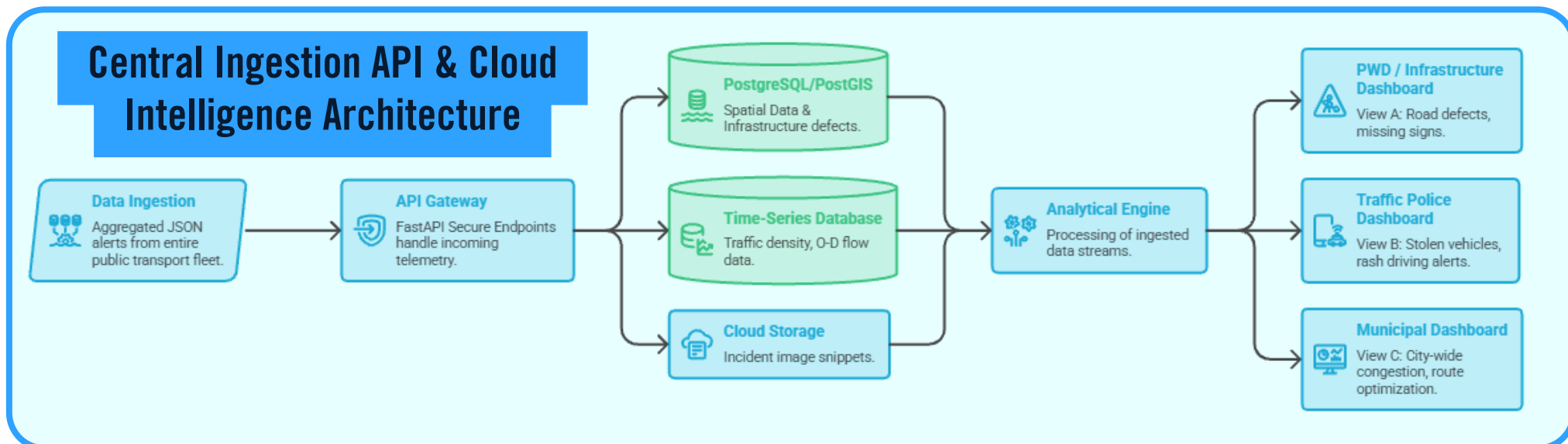
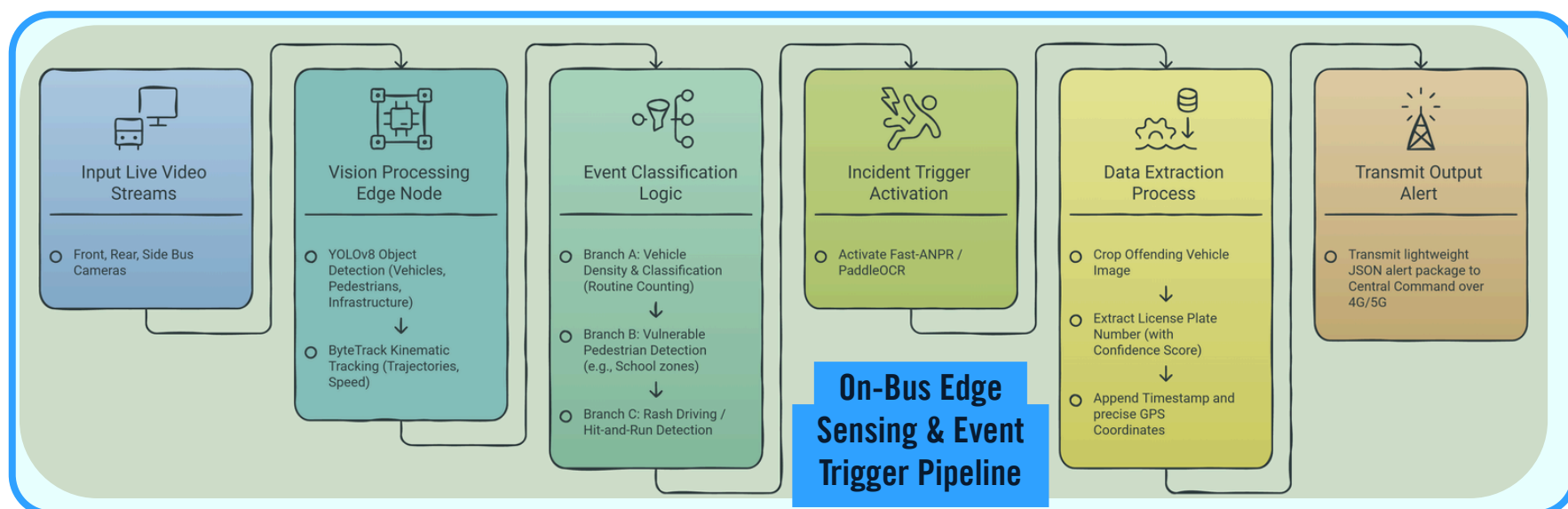
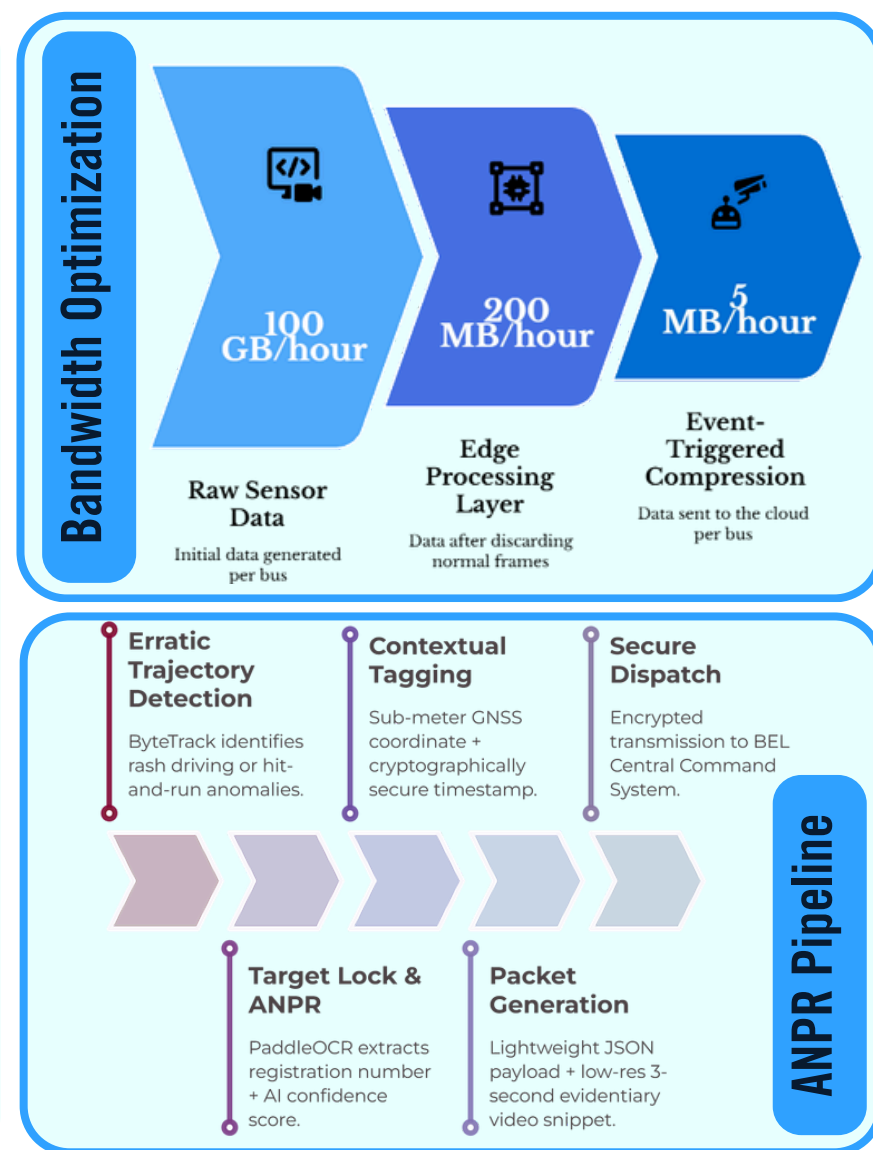
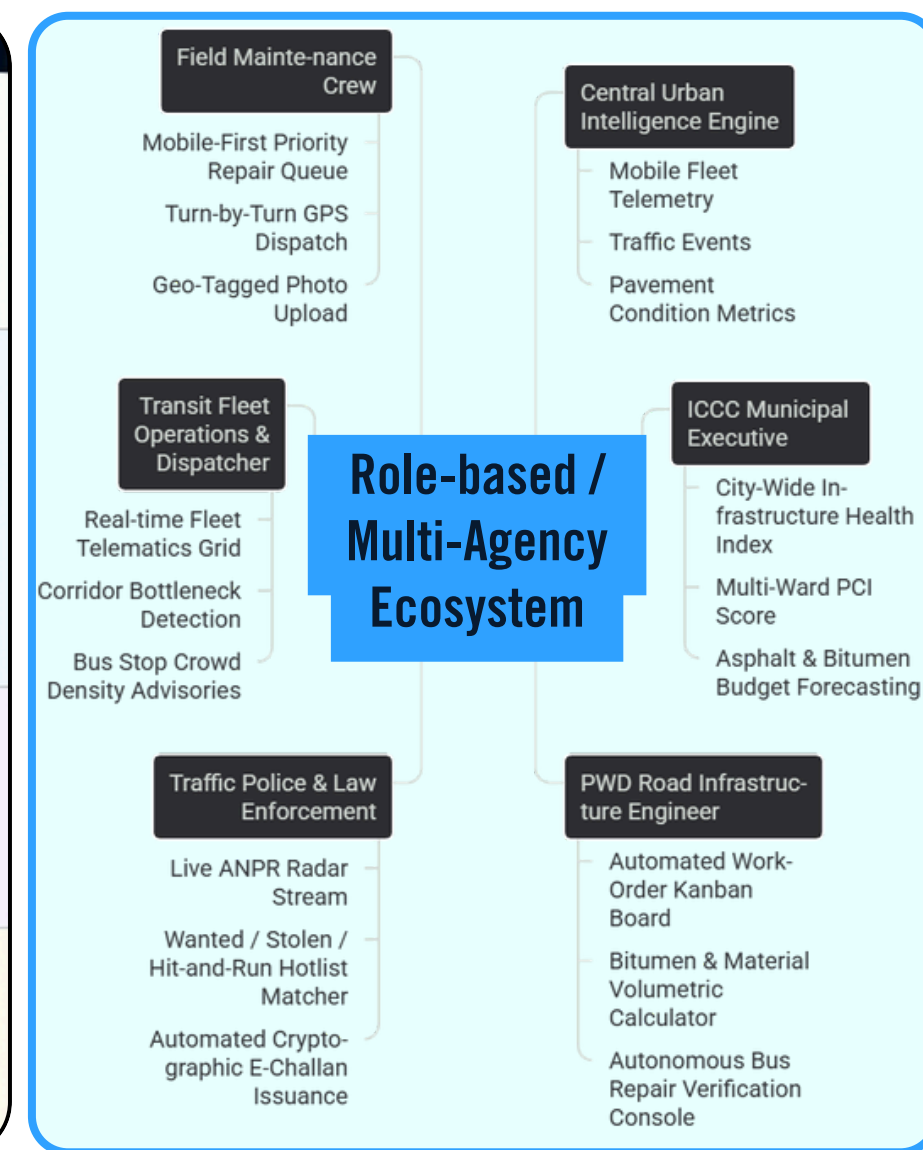
2. How it addresses the problem

Metric	STRATA Edge-AI Mesh	Mobile LiDAR Vans	Static CCTV Network
City Grid Coverage (%)	95%	25%	12%
Survey Refresh Frequency	98% — Every 15–30 mins	10% — Annual	30% — Fixed-point continuous
CapEx Efficiency (Cost Savings)	92% — <₹62k retrofit	15% — ₹3.5 Cr/van	20% — ₹1.5 Cr/km cabling
Multi-Agency Utility	96% — PWD + Police + ICCC + SRTU	30% — PWD only	40% — Police only
Bandwidth Efficiency	99.8% — 15 MB/day Edge JSON	15% — Offline terabytes	20% — 50–100 GB/day/cam

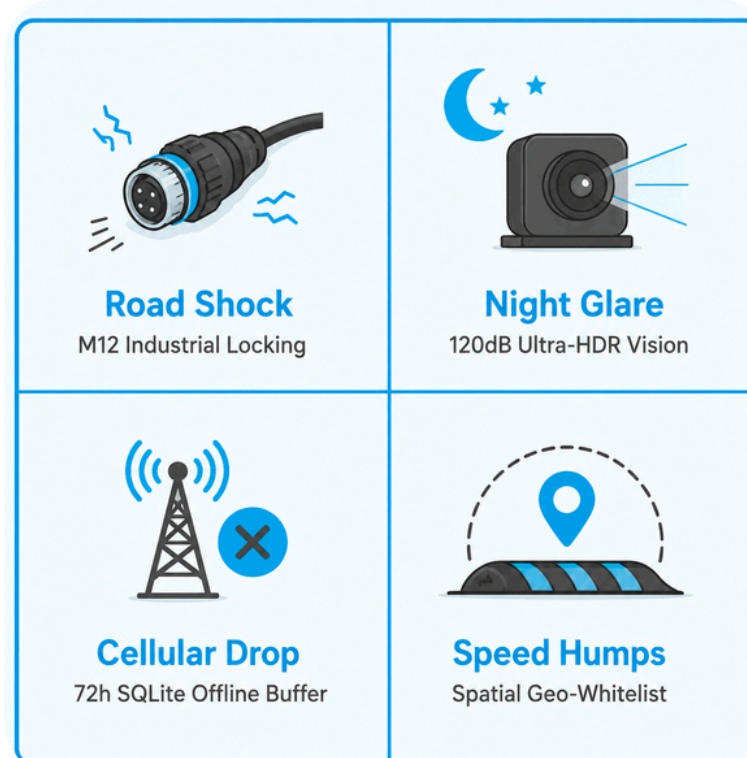
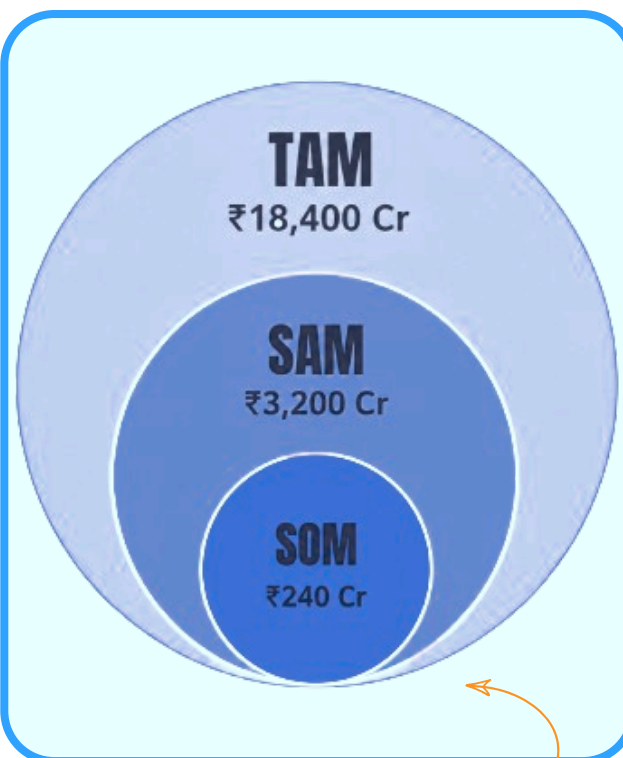
3. Innovation and uniqueness of the solution

- **ImuFusionGate Sensor Fusion:** Eliminates false alarms (*shadows, water glare*). Camera detections are cross-checked against a short IMU vibration window to reject visually convincing but physically unsupported road-defect events.
- **Speed-Proportional Frame Throttling:** Dynamically scales video ingestion (*2 FPS at traffic stops to 30 FPS at cruising speeds*) to optimize compute and power draw.
- **Vulnerable Pedestrian Safety:** Edge vision pipeline detects school children crossing, mid-block jaywalking, and pedestrian bottlenecks in high-density corridors, triggering instant driver alerts.
- **H3 Spatial-Temporal Deduplication:** One defect ≠ 50 alerts, Repeated observations are spatially and temporally clustered so recurring sightings become one persistent infrastructure event using *Uber H3 hexagonal indexing*.
- **Kinematic Tracking:** *C++/Python vision pipeline tracks erratic vehicle trajectories (rash driving) and extracts timestamped ANPR*.

LAYER	TECHNOLOGIES / COMPONENTS	PURPOSE / ROLE
 1. EDGE COMPUTE & SENSORS	 NVIDIA Jetson Orin Nano (8GB, 40 TOPS, 15W TDP)  4x Sony IMX390 Automotive HDR Cameras  6-DOF MPU-6050 IMU  U-Blox NEO-M8N GNSS	- On-board AI inference & edge processing 360° perception: road, traffic, kerb, cabin - Vibration & motion validation (Z-axis spikes) - High-precision positioning & geotagging
 2. EDGE VISION & MODELS	 Python 3.11  Ultralytics YOLOv8  TensorRT 10.x INT8 Quantization  ByteTrack Kinematics  PaddleOCR / Fast-ANPR  GStreamer	- Edge application runtime - Real-time detection of road distress & objects - Optimized, low-latency AI inference - Multi-object tracking & vehicle kinematics - ANPR & text recognition for number plates - Efficient multi-camera video pipeline
 3. CLOUD CORE & SPATIAL ENGINE	 FastAPI (Async Gateway)  Redis Pub/Sub  PostGIS / PostgreSQL  Uber H3 Hexagonal Spatial Indexing (Res 10/11)	- Async APIs for telemetry & command - Real-time event streaming & messaging - Geospatial storage, analytics & querying - Spatial clustering & deduplication of detections
 4. COMMAND DASHBOARDS	 Next.js 15 (React 19, TypeScript)  Tailwind CSS v4  Deck.gl / MapLibre GL 3D Vector Maps  Zustand State Engine	- Modern full-stack React framework - Responsive UI styling & design system 3D maps, heatmaps & spatial visualizations - Lightweight global state management

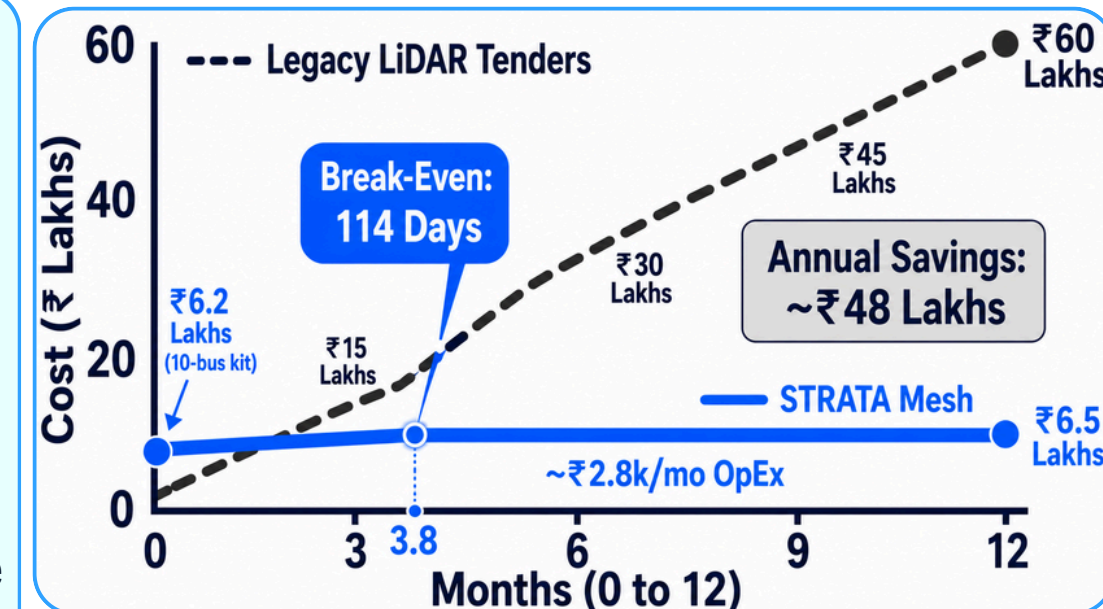
End-to-End Fleet Telemetry & Multi-Camera Dataflow

FEASIBILITY AND VIABILITY



Economic Viability, Cost Structure & B2G Revenue Model

- **CapEx & Fleet Hardware BOM (Per Bus):**
 - Jetson Orin Nano (40 TOPS): ₹42,000 | 4x Sony HDR Cams: ₹12,800
 - 6-DOF IMU + GNSS: ₹2,400 | IP66 Housing & Power: ₹4,500
 - Total Unit BOM: ₹61,700 (vs ₹2.5–₹3.5 Cr LiDAR survey van).
 - Rapid Depot Retrofit: <90 min plug-and-play install per bus.
- **OpEx & Data Viability (>99.8% Savings):**
 - Raw video (~50 GB/day) → Edge telemetry (~15 MB/day/bus).
 - Sustained via standard ₹299/month commercial 4G/5G SIMs.
- **Recurring B2G SaaS Revenue Streams:**
 - Municipal DaaS Tier: ₹2,500 – ₹3,500 / bus / month subscription for live ICCC spatial layers, PCI health map, and PWD dispatch.
 - Autonomous Audit Fee: 0.75% transaction fee per auto-verified contractor road repair ticket.
 - City ROI Payback: Full hardware CapEx recovered in <4 months via eliminated third-party road survey tenders.



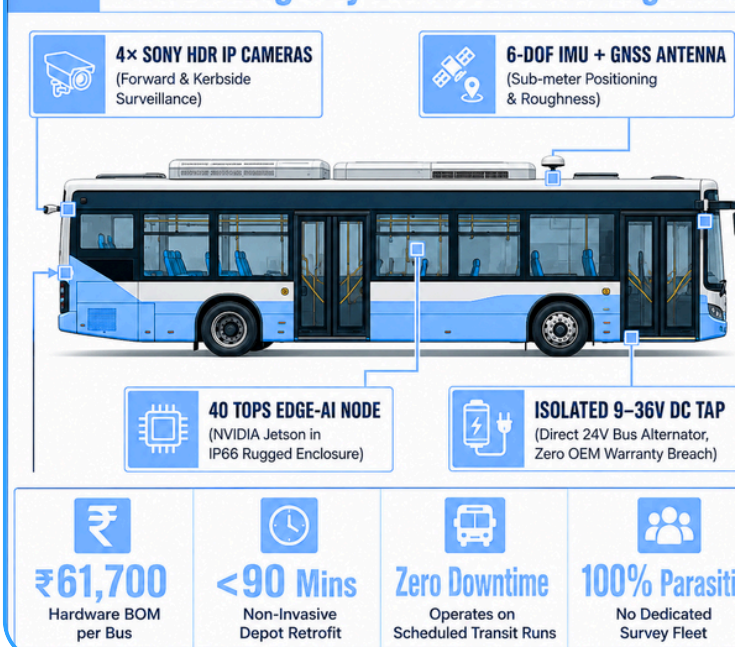
Technical Feasibility & Risk Mitigation Matrix

- **Low Power & Thermal Envelope:**
 - Operates at 12–15W TDP powered directly via isolated 9–36V DC converters from 24V bus alternator circuits; zero battery drain.
 - Passive finned aluminum heat-sink validated up to 60°C ambient temperatures.
- **Vehicle Vibration & Harsh Dynamics:**
 - Risk: Sensor misalignment or connector loosening under heavy road shock.
 - Mitigation: Heavy-duty M12 threaded circular locking connectors with internal silicone vibration damping.
 - Depot Operational SLA: <90-min non-invasive bus retrofit; AIS-140 compliant power isolation preventing vehicle battery drain.
- **Cellular Blackspots & Tunnels:**
 - Risk: 4G/5G signal drops in dense urban corridors.
 - Mitigation: Local SQLite circular ring buffer holds up to 72 hours of offline telemetry with automatic sync and exponential backoff upon reconnection.
- **Adversarial Noise (Speed Breakers):**
 - Risk: False vibration triggers from normal road humps.
 - Mitigation: PostGIS spatial whitelist of authorized municipal speed bumps masks false pothole tickets.

Market Sizing (TAM / SAM / SOM — India First, Then Global)

- **TAM (Total Addressable Market) — ₹18,400 Cr (\$2.2B):**
 - Global Smart Cities Mission & Intelligent Transportation Systems (ITS) road asset inspection market across 10,000+ urban local bodies (ULBs).
- **SAM (Serviceable Addressable Market) — ₹3,200 Cr (\$385M):**
 - 150,000+ State Road Transport Undertaking (SRTU) & municipal city buses across Tier-1/Tier-2 Indian cities (DTC, BEST, BMTCL, CTU).
- **SOM (Serviceable Obtainable Market) — ₹240 Cr (\$29M in 3 Years):**
 - Initial deployment across 4,000 city buses in 12 Smart Cities (MoHUA flagship cities & BEL ICCC integrated hubs).

01 DEPLOYMENT FEASIBILITY: Converting City Buses into Sensing Fleet



ONBOARD RETROFIT HARDWARE KIT



**Municipal Corporations**

Real-time citywide Pavement Condition Index (PCI 0–100) heatmap and data-driven, dynamic budget allocation based on actual road degradation vectors.

**PWD Engineers**

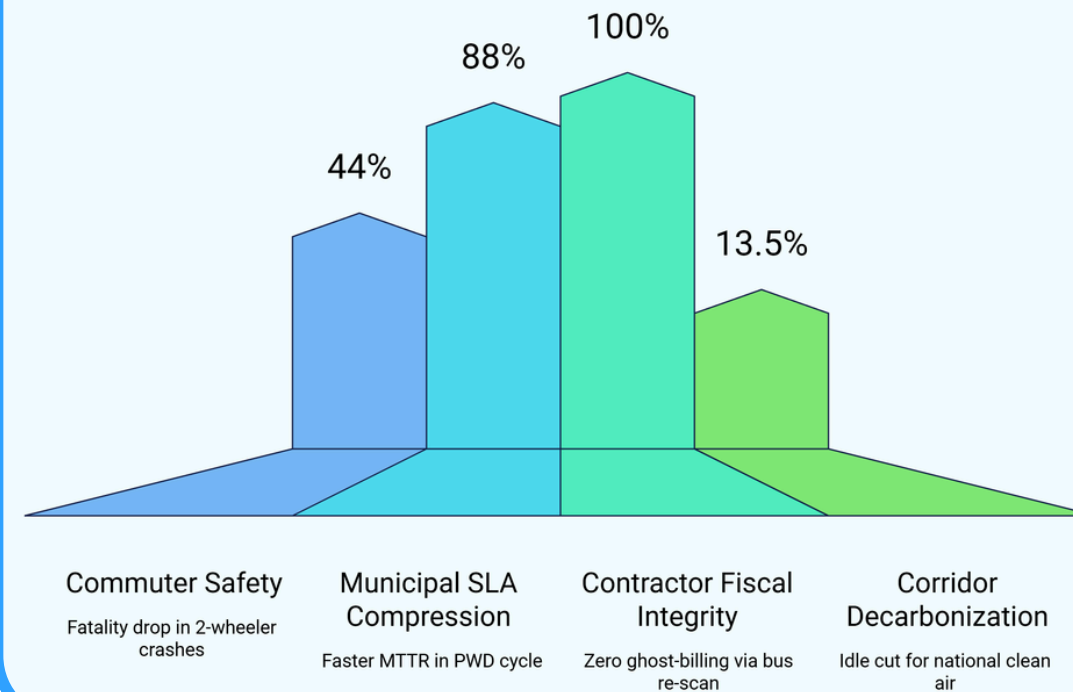
Auto-generated, GPS-pinned work orders with asphalt tonnage calculations and automated pre- and post-repair quality audits.

**Traffic Police**

Fleet-mounted HSRP ANPR radar identifying stolen vehicles, hotlist suspects, and lane violations across camera blind spots.

**Citizens & Commuters**

Pothole-free, smoother arterial roads and faster grievance redressal without filing manual portal complaints.

Performance Metrics of Government Initiatives**Environmental & Sustainability Benefits**

- 12–15% Idling & Decarbonization: Smoother pavement reduces braking-acceleration drag, cutting localized vehicle stop-and-go emissions (NCAP aligned).
- 0 MT Incremental Fleet Carbon: Operates 100% parasitically on scheduled transit routes, avoiding dedicated diesel survey van emissions (~42 kg CO₂/day).
- 45 MT Civil Carbon Offset: Preemptive patching prevents full road resurfacing and sub-base excavation, saving ~45 MT of embodied bitumen emissions per 10 km.

Social Benefits (Commuter Safety)

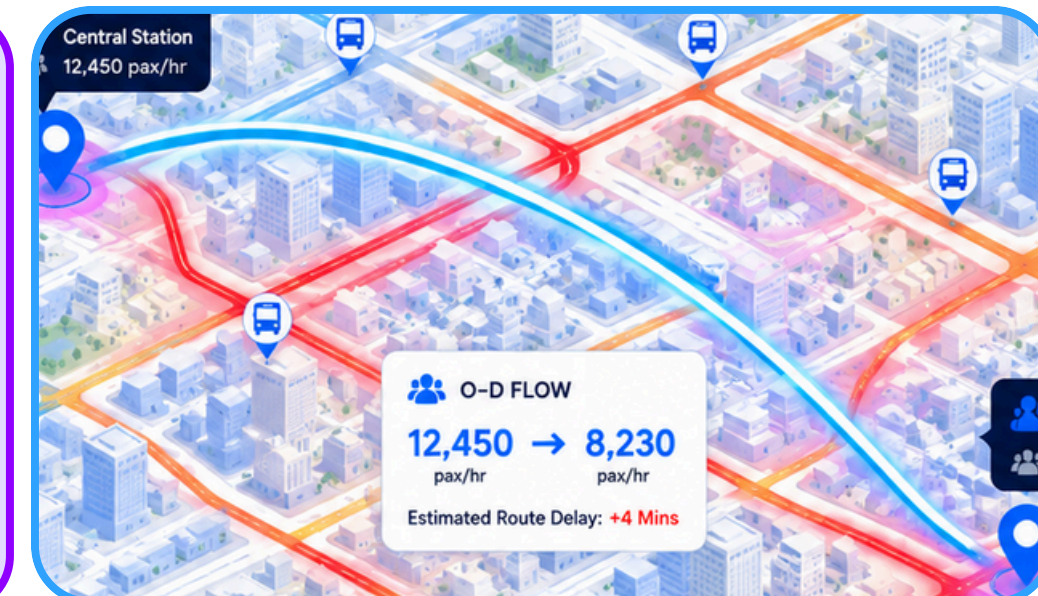
- 44% Two-Wheeler Fatality Mitigation: Targets MoRTH crash baseline (9,400+ annual pothole fatalities) by auto-tagging micro-fissures before fatal crater formation.
- School Zone Safety: Proactively alerts civic authorities to missing crosswalk markings and signs near educational institutions.
- 100% Suburban Inspection Parity: Delivers identical 15–30 min scan frequencies to peripheral low-income bus routes as VIP central avenues.

Economic Impacts

- Zero Ghost-Repair Payouts: Closes procurement leakage by withholding municipal contractor disbursements until subsequent bus runs confirm vertical vibration stays below 1.5g.
- Pavement Structural Life Extension: Timely micro-surfacing halts base-course aggregate failure, reducing recurring multi-year complete road reconstruction CapEx by 35%.
- Survey Tender Disintermediation: Eliminates recurring commercial third-party inspection tenders, releasing municipal funds directly for core public amenities.

Law Enforcement & Administrative Accountability

- Blind-Spot Corridor Enforcement: Mobile bus-mounted ANPR radar extends stolen vehicle and hotlist tracking into outer arterial routes lacking fixed CCTV poles.
- Dedicated BRT Lane Defense: Automatically flags private vehicle encroachments into designated bus rapid transit lanes to preserve public transit right-of-way.
- Indisputable Digital Audit Trail: Cryptographically timestamped optical before-and-after evidence snippets eliminate citizen disputes and hold up against judicial challan appeals.



Reference / Research Work	Relevance to STRATA	Link
AIS-140 – Intelligent Transportation Systems (ITS) for Public Transport Vehicles	Provides the Indian regulatory/ITS foundation for public-transport vehicle tracking, GNSS-based positioning, vehicle telemetry and transport-control integration . Directly relevant to STRATA's bus-mounted sensing node. (Ministry of Road Transport and Highways)	AIS-140 – MoRTH Official PDF
IRC / MoRTH – Road & Pavement Engineering Specifications	Provides the Indian institutional engineering basis for pavement works, road maintenance, defect assessment and repair workflows used by STRATA's PWD module.	MoRTH Official Portal
ASTM D6433-24 – Pavement Condition Index (PCI) Surveys	Establishes the internationally recognized PCI methodology for pavement surface-condition assessment , giving STRATA's 0–100 road-health scoring an established engineering foundation. (ASTM Store)	ASTM D6433-24 – Official Standard
IEEE – Sensor Fusion Research / IEEE Transactions on Intelligent Transportation Systems	Provides research grounding for multi-sensor fusion in intelligent transportation systems . STRATA applies the principle by correlating camera-based road-defect detections with high-frequency IMU measurements to improve confidence and suppress optical false positives.	IEEE Xplore – Intelligent Transportation Systems
Uber H3 – Hierarchical Hexagonal Geospatial Indexing	Provides the spatial-computing foundation for GNSS-to-cell indexing, spatial aggregation and multi-bus defect clustering/deduplication . H3 is explicitly hierarchical and supports multi-resolution geospatial analysis. (H3 Geo)	H3 Official Documentation

STRATA IS FEASIBLE. SCALABLE. COST-EFFECTIVE. READY FOR REAL-WORLD DEPLOYMENT.

BUILDING SMARTER CITIES WITH EVERY BUS.

DETECTION PERFORMANCE (Per Class)

Class	Precision	Recall	F1-Score
Pothole	94.2%	92.1%	93.1%
Waterlogging	93.6%	90.3%	91.9%
Divider Damage	91.7%	89.4%	90.5%
Zebra Crossing	96.5%	95.1%	95.8%
OVERALL (All Classes)	93.7%	91.7%	92.7%

Model: YOLOv8n (Custom Trained) | Confidence Threshold: 0.4

SENSOR FUSION VALIDATION

Camera Only + IMU (Vibration) = Camera + IMU

False Positives: 18.7% → 4.2%

77.6% Reduction in False Positives

Best for distinguishing Potholes vs Speed Breakers

EDGE PERFORMANCE (Jetson Orin NX)

Avg. Inference Time (Per Frame)	68 ms	Max Throughput (1080p)	14 FPS
Power Consumption	18.6 W	Operating Temp.	32°C - 56°C
DATA COLLECTED	48+ Hours, 1.2+ Lakh Frames	EVENTS DETECTED	3,842 Total Events, 4 Defect Classes

Primary Research & Voice of Market

- Municipal Corporation / PWD (Field Engineers):**
 - Ground Truth: 80% of citizen complaints on Swachhata/CPGRAMS lack accurate GPS or depth metrics, causing 3–5 week re-inspection delays.
 - Validation: PWD confirmed automated volume estimation (Area × Depth × 2.4 MT/m³) eliminates contractor over-billing on asphalt tenders.
- State Transit Authorities (DTC / CTU Depot Managers):**
 - Operational Constraint: Bus fleets reject any system modifying 24V wiring looms or causing service route downtime.
 - Validation: Solved via <90-min overnight depot retrofit and isolated 9–36V DC taps preserving OEM electrical warranties.
- Traffic Police & Enforcement Officials:**
 - Ground Truth: Static CCTV misses rash driving, wrong-way movement, and hotlist suspects in blind spots. Moving bus ANPR enables continuous mobile corridor tracking.

Industry Outlook (2026–2030)

- MoHUA Smart Cities Mission 2.0:** Over 100+ ULBs are transitioning from passive CCTV monitoring to AI-driven automated municipal operations.
- National Clean Air Programme (NCAP):** Congestion hotspot identification through bus fleet tracking directly supports urban corridor emission cuts.
- MoRTH Digital Road Asset Mandate:** Mandates digital Pavement Condition Index (PCI) data for performance-based road maintenance contracts (OPRC).
- PM-eBus Sewa Fleet Expansion:** 50,000+ electric buses deploying across Tier-2/Tier-3 cities provide a standardized 24V host infrastructure for rapid STRATA rollout.

